25900298 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECND200 - ADVANCED NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. BMR, SDA and ..... is used to determine total energy requirement. [K] / [CO1]

   a). BMI b). Activity cost

   c). BMR d). REE

2. The hormone primarily responsible for water reabsorption in kidneys [U] / [CO1]

   a). Insulin b). Antidiuretic hormone

   c). Aldosterone d). Cortisol

3. The presence of fat in intestine stimulate the hormone ..... [U] / [CO2]

   a). Ghrelin b). Amylase

   c). Lipase d). Cholecystokinin

4. The net dietary protein value expressed as percentage of total calories. [U] / [CO2]

   a). PER b). NDP Cal%

   c). NPU d). NPR

5. Serum level of vitamin E below …….is state of toxicity. [U] / [CO3]

a). 3.5 mg/dl b). 5.5 mg/dl

c). 6 mg/dl d). 35 mg/dl

6. Bowing of leg is seen in deficiency of ……. [U] / [CO3]

a). Vitamin D b). Vitamin B

c). Vitamin A d). Vitamin E

7. Which transport proteins carry vitamin B12 in blood [K] / [CO3]

a). Transcobalamins I, II, III b). Albumin

c). Globulin d). Haemoglobin

8. Which type of cretinism is linked to goitrogens consumption [U] / [CO4]

a). Neurological b). Myxoedematous

c). Both d). None

9. Which of the following hinders selenium absorption [K] / [CO4]

a). Phytates b). Glutathione

c). Vitamins C, A, E d). Selenate

10. Bioavailability of zinc from ……..is high [U] / [CO4]

a). Organ meats b). Human milk

c). Cow's milk d). Fruits

[1x10 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **2** marks

11. Define calorimetry. [K] / [CO1]

12. State how water act as a part of structure [U] / [CO1]

13. Discuss about gums. [U] / [CO2]

14. Discuss the haematological manifestations in pyridoxine deficiency [U] / [CO3]

15. Discuss how vitamin A deficiency leads to infection. [U] / [CO3]

16. Describe the absorption of magnesium [K] / [CO4]

17. Indicate the relationship between ageing and calcium absorption [U] / [CO4]

[2x5 = 10]

Part C

Short Essay

Answer any **6** questions

Each question carries **5** marks

18. Explain the functioning of Benedict Roth Respirometer. [U] / [CO1]

19. Discuss the role of fibre in preventing colon cancer. [U] / [CO2]

20. Explain the digestion, absorption and transport of protein. [U] / [CO2]

21. Discuss the clinical features of pellagra [U] / [CO3]

22. Explain scurvy- causes and clinical manifestations in adults [U] / [CO3]

23. Explain the causes and clinical features of thiamin deficiency [U] / [CO3]

24. Differentiate skeletal fluorosis and dental fluorosis [U] / [CO4]

25. Explain the process of absorption and metabolism of phosphorus [U] / [CO4]

[5x6 = 30]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

26. Explain TER and its components. [U] / [CO1]

27. Explain the digestion, absorption and transport of carbohydrates. [U] / [CO2]

28. Explain folate deficiency, causes, clinical features and its effect on foetal development [U] / [CO3]

29. Describe the absorption, transport, storage, and metabolism of iron in the human body [U] / [CO4]

[10x2 = 20]